

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained **Python** and **SQL** components for data ingestion and integration workflows, processing approximately 1,500 daily transactions.
- Implemented and debugged AI/ML features based on guidance from senior architects, contributing to pilot projects for enterprise-scale solutions.
- Collaborated with data engineers and platform teams to ensure AI workloads could access and process relevant data, optimizing data retrieval for downstream tasks.
- Assisted in the creation and maintenance of reusable deployment pipelines for AI models, supporting automated testing and deployments using **GitHub Actions**.
- Integrated services such as **Document AI** and **Microsoft Copilot extensions** by developing connector logic and APIs to bridge existing enterprise systems.
- Contributed to **telemetry**, **logging**, and **performance optimization** efforts for AI workloads, identifying and resolving bottlenecks in processing chains.
- Containerized AI services using **Docker** and deployed them to a **Kubernetes** cluster, managing configurations and resource allocation for stable operation.

SKILLS

Programming Languages – Python, SQL

AI/ML Frameworks – LangChain, Semantic Kernel, Streamlit

Cloud AI Platforms – Snowflake (Snowpark, Document AI, Cortex), Azure AI (OpenAI, AI Foundry, Azure ML)

Data & Orchestration – Vector Databases, Airflow, Azure Data Factory

Software Engineering – APIs, SDKs, CI/CD, GitHub Actions, Azure DevOps, Docker, Kubernetes

PROJECTS

Enterprise AI Integration Platform

- Developed a **Python**-based integration layer using **FastAPI** to connect legacy enterprise systems with new AI services, handling approximately 500 API calls per hour.
- Implemented **RAG (Retrieval-Augmented Generation)** patterns for a document analysis service, enabling the AI to query a **vector database** and provide contextually relevant answers.
- Designed and deployed the integration service on **AWS Lambda** and **API Gateway**, ensuring scalable and cost-effective operation for internal stakeholders.

AI-Powered Customer Support Assistant

- Built a proof-of-concept **AI assistant** using **LangChain** and **OpenAI** models to automate responses to common customer inquiries, reducing manual handling by an estimated 15%.
- Developed data ingestion pipelines with **Python** and **Pandas** to process customer interaction logs for model fine-tuning, preparing a dataset of ~10,000 interactions.
- Integrated the assistant's backend with a sample **REST API**, allowing for programmatic interaction and testing of its conversational capabilities.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)